

Learn the role. Solve the world.

A small agent learns geometric roles through interaction, then uses that evidence in unseen procedural worlds.

639/640 held-out worlds solved	33.62% fewer actions than cold	18.317 actions saved per world	0 external API calls
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The result

In the frozen v4 run, five separately trained models solved 639 of 640 held-out worlds after 16 training examples per family. They averaged 36.17 actions per world, versus 54.49 for the same fixed priors without retained evidence. The paired difference was 18.317 actions (95% stratified paired-bootstrap interval 16.461 to 20.056).

Condition	Success	Mean actions	Purpose
Trained, 16 examples/family	639/640 (99.84%)	36.17	Retained role evidence
One example/family	625/640 (97.66%)	45.79	Low-data comparison
Cold model	621/640 (97.03%)	54.49	Same priors, no retained evidence
Random actions	35/640 (5.47%)	125.11	Unguided reference

The world is a 13×13 grid with opaque tile IDs, remapped actions, twelve decoys, and up to eighteen obstacles. Training, held-out, and shift-probe seeds and token ranges are disjoint.

A result with a visible boundary

Under reversed key/exit roles, the trained model solves 137/160 worlds at 92.31 mean actions; cold solves 156/160 at 59.36. The transferred rule causes about 32.95 extra actions per world. This negative transfer is the next research problem, not a result hidden from the proposal.

Scope: synthetic grid worlds, five declared geometric features, and one independently trained model per family. This does not demonstrate unknown-family discovery, raw visual perception, general intelligence, Rust integration, or ARC-AGI performance.

One learner. Five families. Safer transfer.

The \$10,000 request funds the unresolved step exposed by the result: a shared learner that can select and revise role models from interaction without receiving a task-family label.

Six-week plan

Period	Work	Evidence
Week 1	Freeze the v4 reference	Clean-run reproduction, protocol, and checksums
Weeks 2-3	Build a shared learner	No family label; interaction traces and development benchmark
Weeks 4-5	Test model selection and shift recovery	New frozen split; paired cold, reset, and adaptive comparisons
Week 6	Release code and report	One-command reproduction, results, and failures

Predeclared evaluation gate

Keep the v4 reference unchanged. Test at least 1,000 new held-out worlds, keep family labels and hidden state out of the policy, report every family, and compare against paired cold, reset, and random baselines. A Rust port starts only after the shared learner passes; if it does not, publish the failure and revise the claim.

Proposed use of funds

Use	Amount	Deliverable
Researcher time	\$7,000	Shared learner and experiments
CPU evaluation and storage	\$2,000	Frozen splits, ablations, preserved results
Documentation and release	\$1,000	Reproduction guide and technical report
Total	\$10,000	Open research release

The current benchmark uses the Python standard library and makes zero external API calls. Its recorded full-run evaluation time was 175 seconds; hardware affects runtime.

Reproduce the current evidence

Showcase: `bash tools/run_open_intelligence_showcase.sh`

Full benchmark: `bash tools/run_open_intelligence_demo.sh`

Repository: github.com/lipcsiklaszlo5-lgtm/Athlesia

Independent research. Open code. Results tied to a frozen evaluator.

Benchmark: athlesia-open-intelligence-core-v4 | Worlds: 640 held-out + 160 shift probe | Prepared 24 September 2026